IT Internship Course Outline

A structured 30-day learning programme for B.Tech CSE Interns

Designed & Maintained By

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IT Internship Course Outline

HURL Sindri – IT Department

Duration: 20-25 Working Days | Target: B.Tech CSE/IT (4th Year Intern)

Scope: Sandboxed learning environment – no access to live plant or sensitive systems

Internship Objectives

- Provide structured exposure to IT operations in an industrial/plant setting
- Build practical understanding of networking, systems and enterprise IT tools
- Develop awareness of plant IT security considerations and OT/IT environments
- Produce documented outputs that demonstrate applied learning

Program Structure at a Glance

Phase	Days	Focus Area	Key Output
Phase 1	Days 1–7	Foundations & Networking	Architecture sketch, LAN report, VLAN diagram
Phase 2	Days 8–15	Plant-Specific Systems	CCTV zone map, SAP Exploration Report
Phase 3	Days 16–23	Infrastructure & Support	AD task report, Troubleshooting Cheat Sheet
Phase 4	Days 24–30	Security & Closure	Security document, Final Internship Report

Daily Schedule – Detailed Breakdown

■ PHASE 1: FOUNDATIONS & ORIENTATION

Day 1 – IT in a Fertilizer Plant – The Big Picture

Topics:

- Role of IT in plant operations – how IT enables production, safety, and administration
- Corporate IT vs. Plant IT – differences in priorities, criticality, and risk
- Overview of IT systems at HURL: SAP, Network Infrastructure, CCTV, Biometric Access, End-User Support
- Introduction to IT governance and policies relevant to internship

Task:

Draw a basic diagram of 'IT Systems in a Fertilizer Plant' – showing all major domains and how they interconnect

Expected Output:

→ **1-page architecture sketch (hand-drawn or digital)**

Day 2-3 – Networking Fundamentals + OT/IT Convergence

Topics:

- Networking basics (structured revision): IP addressing, Subnetting, VLANs
- LAN vs. WAN vs. MPLS – use cases in enterprise and plant environments
- Devices: Switch, Router, Firewall – roles and placement
- OT/IT Convergence: What separates Operational Technology (OT) networks from IT networks
- Plant-specific networking: Why plant networks include segments for PLCs, SCADA, DCS – and why these are isolated
- Concept of air-gapping and DMZ in industrial environments

⚠ Controlled Exposure:

Sanitized version of plant network diagram – for reference and learning only. No access to live systems.

Task:

Identify and label all major components visible in the sanitized network diagram. Annotate which segments are IT and which are OT.

Expected Output:

→ **Annotated diagram with written explanation**

Day 4-5 – Practical Networking – Hands-On

Topics:

- IP configuration in Windows (static and dynamic)
- Key diagnostic commands: ping, ipconfig, tracert, nslookup
- File sharing over LAN – permissions and mapping
- Why plant IT avoids ad-hoc file sharing – policy context

Hands-On Exercise:

Connect two laptops via a network switch (isolated setup – no plant network access). Configure IPs manually, enable file sharing, verify connectivity.

Expected Output:

→ **Written report: 'How Devices Communicate in a LAN' – include screenshots and observations**

Day 6-7 – VLANs in a Plant Environment

Topics:

- What is a VLAN and why segmentation matters
- Plant-specific VLAN use cases: Admin / Plant Floor / CCTV / Guest
- How VLANs relate to security and OT isolation (connecting back to Day 2)
- Trunk ports, access ports – conceptual overview

Demo (Supervised):

VLAN configuration screenshots shown by supervisor – no live switch access

Task:

Design a conceptual VLAN map for a fertilizer plant. Must include at minimum: VLAN 10 – Admin, VLAN 20 – Plant/OT, VLAN 30 – CCTV, VLAN 40 – Guest/Vendor

Expected Output:

→ **VLAN diagram with a short written justification for each segment**

PHASE 2: PLANT-SPECIFIC SYSTEMS

Day 8-10 – CCTV & Biometric Systems

Topics:

- Role of CCTV in plant security and operations monitoring
- Types of cameras: IP cameras, PTZ, fixed – where each is used
- NVR/DVR setup – storage and retention concepts
- Biometric access control: how fingerprint/card-based systems work
- Integration of CCTV and biometric data with IT infrastructure
- Policy: who has access to footage, audit trails, privacy considerations

Demo (Supervised):
Supervised walkthrough of CCTV interface (viewing only, no configuration access)

Task:
Map the CCTV zones of a hypothetical plant layout. Identify blind spots. Justify camera placements.

Expected Output:
→ **Zone map with written justification**

Day 10-14 – SAP at HURL – Awareness & Structured Exploration

Topics:

- What is an ERP? Why plants use SAP
- Modules in use at HURL: MM (Materials Management), FI (Finance), PM (Plant Maintenance)
- How modules are interconnected – e.g., a purchase order in MM triggering a posting in FI
- Key terminology: Material Code, Transaction Code (T-code), Cost Center, Work Order

Demo (Supervised):
Supervised SAP interface walkthrough – read-only access, no data entry

Task:
Material Procurement to billing workflow with T-Codes

Expected Output:
→ **SAP Exploration Report: 1 page covering all 3 T-codes with screenshots**

PHASE 3: INFRASTRUCTURE & SUPPORT

Day 15-18 – Windows Server & Active Directory Basics

Topics:

- What is Active Directory and why organisations use it
- Key concepts: Domain, Domain Controller, Users, Computers, Groups
- Organisational Units (OUs) – what they are and how IT uses them to organise users and machines

Demo (Supervised):

Adding new user, setting password, configuring basic account properties, adding computer to domain

Task:

Given a hypothetical scenario — "5 new employees have joined across Admin, Plant Ops, and IT. One employee has resigned." — perform or document the step-by-step AD actions required for each case.

Expected Output:

→ **AD Task Report – a simple table listing each scenario, action taken, result observed**

Day 19-23 – Remote User Support & SLA Basics

Topics:

- Remote desktop tools — TightVNC and AnyDesk: how they work, when to use which
- Basic PC troubleshooting — slow system, not booting, display issues, peripheral not detected, OS installation
- Basic network troubleshooting — no connectivity, IP conflicts, DNS issues (ping, ipconfig, tracert)
- Basic printer troubleshooting — offline printer, print queue stuck, driver issues

Task:

For each of the four areas above, document a simple first-response checklist — what you check first, second, and when you escalate.

Expected Output:

→ **A one-page "First Response Troubleshooting Cheat Sheet" covering PC, network, printers and remote tools**

PHASE 4: SECURITY & CLOSURE

Day 24-30 – IT Security in a Plant Environment

Topics:

- Why plant IT security is uniquely critical – consequences of a breach go beyond data loss
- Common threats: USB-based attacks, phishing, unauthorized access, insider threats
- OT-specific risks: what happens if the OT network is compromised
- Best practices: endpoint security, patch management, access control
- Physical security as part of IT security: server room access, equipment handling

Task:

Create a 'IT Security Do's & Don'ts' document for plant employees (1 page, A4). Must cover: USB usage, password hygiene, phishing awareness, physical access.

Expected Output:

→ **Printable 1-pager suitable for display on notice boards**

Day 30 Onwards – Internship Report – Preparation & Submission

Topics:

- Structure and review all outputs from Days 1–30
- Compile into a structured internship report

Report must include the following sections:

- Introduction: Role of IT in HURL Sindri
- Learnings: One paragraph per major topic area
- Hands-on work: Summary of tasks completed
- Key takeaways: What was new, what was surprising
- Suggestions: One improvement idea for IT processes (optional)

Expected Output:

→ **Final Internship Report – submitted to reporting manager**

Ground Rules & Sandboxing Policy

- Intern has NO access to live plant networks, OT systems, or production servers
- All demos are supervisor-led; intern observes unless explicitly told otherwise
- Network exercises use an isolated lab setup only
- No data, configurations, or screenshots may leave the premises without approval
- All output documents are reviewed by the reporting manager before submission